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Detecting Heart Abnormality Using ECG with CART and Random Forest

Mini Project Report — Reproduction and Extension

1. Problem Statement

Electrocardiogram (ECG) signals contain clinically useful information for identifying cardiac abnormalities. The objective of this mini-project is to reproduce the main machine-learning approach reported in the reference study for heart-abnormality detection and to extend the work using one additional model. The project therefore evaluates a CART decision tree on the UCI Arrhythmia dataset and compares it with a Random Forest extension using the same evaluation setting.

2. Dataset

The UCI Arrhythmia dataset was used for this project. It contains 452 patient records and 279 original attributes. The task is multiclass classification involving normal and abnormal cardiac conditions. The reference study also uses the UCI Arrhythmia dataset. Missing entries in the dataset are represented by the character '?'.

The dataset was inspected before modelling. Feature 14 was removed because it contains a very large proportion of missing values in the reference dataset. The remaining data was retained rather than deleting a large number of patient records.

3. Data Preprocessing

The '?' symbols were first converted into missing values. Feature 14 was removed. Remaining numerical missing values were handled using median imputation. After preprocessing, the feature matrix contained 452 records and 278 input features.

For model evaluation, a stratified 70:30 train-test split was used with random_state = 42. This produced 316 training samples and 136 testing samples. Stratification was used to maintain the class distribution between the training and testing sets.

4. Methodology

CART — Reference Model: CART was implemented using Python's DecisionTreeClassifier with the Gini criterion and random_state = 42. The trained model was evaluated using accuracy, precision, recall and F1-score. A confusion matrix and ten-fold cross-validation were also used for evaluation.

Random Forest — Extension Model: Random Forest was selected as the additional model because it combines predictions from multiple decision trees and can reduce the limitations of relying on a single tree. It was trained using the same preprocessed data, train-test split and evaluation metrics. The implementation used 100 trees and random_state = 42.

5. Experimental Results

The following results were obtained on the 30% held-out test set:

Model	Accuracy	Precision	Recall	F1-Score
CART	69.12%	67.46%	69.12%	67.80%
Random Forest	75.00%	68.10%	75.00%	69.47%

The CART model achieved an accuracy of 69.12%, with precision of 67.46%, recall of 69.12% and F1-score of 67.80%. The ten-fold cross-validation performed during the project produced a mean accuracy of 64.79% with a standard deviation of 7.18%.

Random Forest achieved 75.00% accuracy, 68.10% precision, 75.00% recall and 69.47% F1-score. Compared with CART, Random Forest improved test accuracy by 5.88 percentage points and also produced higher recall and F1-score.

6. Reproduction Comparison with the Reference

The reference paper reports approximately 80% classification accuracy for its CART experiment. Our reproduced CART result is 69.12%. The difference is reported transparently because the reference study and this implementation do not use exactly the same experimental configuration. In particular, the reference uses the R rpart implementation with complexity control and pruning, whereas this project uses Python's DecisionTreeClassifier with the Gini criterion and random_state = 42. Differences in implementation, pruning/complexity settings, train-test split, preprocessing details, randomization and library versions can therefore affect the final accuracy.

The reproduction should consequently be interpreted as a close implementation of the reference workflow rather than an exact numerical replication of every experimental setting. The same UCI Arrhythmia dataset and the main CART-based classification idea were retained.

7. Model Comparison and Discussion

Random Forest performed better than CART on the held-out test set. Its accuracy was 75.00% compared with 69.12% for CART. The improvement is 5.88 percentage points. Random Forest also achieved a higher F1-score (69.47% versus 67.80%) and higher recall (75.00% versus 69.12%). This suggests that, under the experimental configuration used in this project, the ensemble model was more effective than a single decision tree.

However, the result does not mean that Random Forest will always outperform CART. Performance depends on the dataset, preprocessing, hyperparameters and train-test split. The comparison here is specifically for the UCI Arrhythmia dataset and the experimental setup used in this mini-project.

8. Conclusion

The project successfully implemented the reference CART-based ECG classification workflow and extended it with Random Forest. The complete pipeline included dataset preparation, missing-value handling, feature removal, median imputation, stratified train-test splitting, model training and evaluation. CART obtained 69.12% test accuracy, while the additional Random Forest model obtained 75.00%. Thus, Random Forest performed better in our experiment. The difference between the paper's reported accuracy and our reproduction was documented rather than hidden, satisfying the reproduction requirement.

Keywords: ECG, Arrhythmia, CART, Decision Tree, Random Forest, UCI Arrhythmia, Classification